

Section I: Data Structures [15 Marks]

Q 1) Write a menu driven program to perform the following operations on stack
i. Accept a postfix expression. ii. Evaluation of postfix expression

Answer:

```
#include <stdio.h>
#include <ctype.h>
#define MAX 100

int stack[MAX], top = -1;

void push(int x)
{
    if(top == MAX - 1)
        printf("Stack Overflow\n");
    else
        stack[++top] = x;
}

int pop()
{
    if(top == -1)
    {
        printf("Stack Underflow\n");
        return 0;
    }
    return stack[top--];
}

int evaluate(char postfix[])
{
    int i, a, b, result = 0;
    char c;

    top = -1;
    for(i = 0; postfix[i]
    != '\0'; i++)
    {
        c =
        postfix[i];

        if(isdigit(c))
            push(c -
            '0');
        else
        {
            b =
            pop();
            a = pop();
            switch(c)
            {
                case '+':
                    result = a + b; break;
                case '-':
                    result = a - b; break;
                case '*':
```

```

result = a * b; break;
result = a / b; break;
    }
push(result);
    }
return pop();
}

int main()
{
    char postfix[MAX];
    int ch, accepted = 0;

    do
    {
        printf("\n1. Accept Postfix Expression\n");
        printf("2. Evaluate Postfix Expression\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &ch);

        switch(ch)
        {
            case
            1:
                printf("Enter postfix expression: ");
                scanf("%s", postfix);
                accepted = 1;
                break;
            case 2:
                if(accepted)
                    printf("Result
= %d\n", evaluate(postfix));
                else
                    printf("Accept the expression first\n");
                break;
            case 3:
                break;
            default:
                printf("Invalid choice\n");
        }
    } while(ch != 3);

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q2) Consider the following database (Primary keys are underlined)

Employee (empno, emp_name, city, designation, salary)

Project (projno, proj_name, status, start_date)

There exists a Many to One relationship between Employee and Project.

Constraint: status Should be “Complete”, “In progress”.

Q 2(A)

A Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Project(  
projno INT PRIMARY KEY,  
proj_name VARCHAR(50),  
status VARCHAR(15)  
CHECK(status IN ('Complete','In progress')),  
start_date DATE  
);
```

```
CREATE TABLE Employee(  
empno INT PRIMARY KEY,  
emp_name VARCHAR(50),  
city VARCHAR(50),  
designation VARCHAR(50),  
salary NUMERIC(12,2),  
projno INT REFERENCES Project(projno)  
ON DELETE CASCADE  
);
```

```
INSERT INTO Project VALUES  
(10,'Robotics','In progress','2025-04-01'),  
(105,'Website','Complete','2024-01-10'),  
(106,'Mobile App','In progress','2025-04-01');
```

```
INSERT INTO Employee VALUES  
(1,'Amit','Pune','Developer',40000,105),  
(2,'Neha','Mumbai','Tester',30000,10),  
(3,'Raj','Pune','Developer',45000,10),  
(4,'Sita','Nashik','Analyst',35000,10),  
(5,'Ravi','Pune','Developer',38000,10);
```

Q 2(B)

B Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Find the names of all Employees working on project no 105.

Answer:

```
SELECT emp_name FROM Employee WHERE  
projno = 105;
```

ii. Count the number of Projects which are “In progress”.

Answer:

```
SELECT COUNT(*) FROM Project WHERE  
status = 'In progress';
```

iii. Change start date of project “Robotics” to 15/12/2022.

Answer:

```
UPDATE Project  
SET start_date = '2022-12-15'  
WHERE proj_name = 'Robotics';
```

iv. Increase the salaries of all employees working on project 10 by 5%.

Q 2(C)

C. Write a query to display the project names on which more than 3 employees are working. [4 marks]

Answer:

```
SELECT P.proj_name  
FROM Project P  
JOIN Employee E ON P.projno = E.projno  
GROUP BY P.projno, P.proj_name  
HAVING COUNT(*) > 3;
```

Q 3) Viva [5 marks]